CandiVue AI — Detailed System Workflow

1. Prepare the Job

Hiring Assistant (LangGraph 1)

- **1 — WEB APP + API — Open the job chat**
  POST /api/jobs, then messages to /api/assistant/jobs/{id}
- **2 — HIRING ASSISTANT — Describe the role**
  LLM asks questions, then save_job_profile + weights
  - ✨ LLM (Gemini) — chat + tool calls
- **3 — HIRING ASSISTANT · OPTIONAL — Search references**
  rag_search when useful; also form link + closing date
  - ✨ Embedding model — Ollama, local
  - references ↔ Vector store
- **4 — HIRING ASSISTANT — Write the job description**
  write_job_description: 1 LLM request, draft card
  - ✨ LLM (Gemini) — JD writer
- **5 — HUMAN REVIEW · INTERRUPT — Approve, then "Post?"**
  approve / edit / reject, then Post / Don't post
  - rejected / edited → 4
- **6 — HIRING ASSISTANT — Post the job**
  post_to_job_board with the Google Form link
  - Job board (Blogger) — via MCP

Job posted with the Google Form link

2. Collect and Screen CVs

CV Collector + LangGraphs 2 and 5

- **7 — CV COLLECTOR SERVICE — Collect applications**
  every 5 min: form answers + CV files, read-only
  - Google Form + Drive — Sheets + Drive API
  - CV files ↔ CV folder
- **8 — WEB APP + API — Start CV screening**
  Screen CVs button, POST /api/jobs/{id}/screenings
- **9 — CV SCREENING (PARALLEL) — Read each CV**
  text, OCR if scanned, parse, must-have gate (4 at a time)
  - OCR (Tesseract) — scanned CVs
  - ✨ AI checks — Gemini, else local
- **10 — CV SCREENING — Score each CV**
  rules + local embeddings: 4 scores, weighted 0-100
  - ✨ Embedding model — Ollama, local
- **11 — CV SCREENING — Rank and save**
  ranking with reasons; GET .../screenings/latest
  - scores, ranking ↔ Database
- **12 — TALENT REDISCOVERY · F11 — Check the talent pool**
  past applicants who opted in, same subgraph: suggestions

Ranked list saved

3. Invite Candidates

Hiring Assistant (LangGraph 1)

- **13 — HIRING ASSISTANT — Shortlist**
  update_shortlist: "top 5", "above 75" (applied in code)
  - ✨ LLM (Gemini) — words to criteria
- **14 — HIRING ASSISTANT — Find times**
  find_free_slots, propose_invitations (checked)
  - Calendar — busy times via MCP
  - cancelled: ask again → 13
- **15 — HUMAN REVIEW · INTERRUPT — Confirm invitations**
  candidates, times, email preview; change or cancel
- **16 — HIRING ASSISTANT — Schedule and send**
  session + private link; email (template via RAG) + event
  - Email (Gmail) — via MCP
  - Calendar — via MCP
  - email template ↔ Vector store

At the scheduled time

4. Hold the Interview

Real-time services + room assistant

- **17 — CANDIDATE PAGE — Open the link**
  6-digit code by email, then a token for this interview
  - Email (Gmail) — via MCP
- **18 — IDENTITY CHECK · F7 — Photo check-in**
  3 camera pictures vs. the form photo; result to interviewer
  - ✨ Face models — YuNet + SFace, local
- **19 — MEDIA SERVICE — Waiting room, call**
  "Let in", LiveKit ticket; video, chat, auto-reconnect
  - LiveKit — WebRTC video server
- **20 — HIRING ASSISTANT · ROOM — Interview questions**
  question bank (RAG), questions from the CV, CV summary
  - ✨ LLM (Gemini) — question suggestions
  - question bank ↔ Vector store
- **21 — TRANSCRIPTION SERVICE — Live transcript**
  one stream per speaker, with times; audio not stored
  - Speech-to-text — Deepgram, WebSocket
  - transcript ↔ Database
- **22 — WEB APP + API — End the interview**
  POST /api/interviews/{id}/end starts LangGraph 3

When the interview ends

5. Assess and Decide

LangGraphs 3 and 4

- **23 — Q&A ANALYSIS · LANGGRAPH 3 — Analyse the answers**
  RAG for job + company answers; F8-F10; about 3 AI requests
  - ✨ LLM + embeddings — Gemini, Ollama
  - documents ↔ Vector store
- **24 — FINAL ASSESSMENT · LANGGRAPH 4 — Build the report**
  1 AI request; traceability check (max 2 repairs)
  - ✨ LLM (Gemini) — findings with evidence
  - report ↔ Database
  - changes requested → 23
- **25 — HUMAN REVIEW · INTERRUPT — Review the report**
  approve, or request changes with a comment
- **26 — WEB APP + API — Rank and compare**
  final ranking of approved reports; compare; PDF/HTML
- **27 — WEB APP + API — Selection email**
  editable text, preview, confirm, then send
  - Email (Gmail) — via MCP

Steps: Web App + API (FastAPI, React) · Hiring Assistant chatbot (LangGraph 1) · Human review (LangGraph interrupt) · Background LangGraph workflows (2-5) · Backend services (collector, identity, media, transcriber)

Parts: ✨ AI model — Gemini or local · External service — outside CandiVue · Database · Vector store · CV folder

Database = PostgreSQL (+ LangGraph checkpoints) · Vector store = pgvector · CV folder = Docker volume

Lines: solid = next step; dashed with two heads = data read and written; dashed loop = back after a review.
Data stores are drawn only where a step mainly depends on them; every step also saves its results in the Database (PostgreSQL).
MCP servers (job board, email, calendar) hold their own credentials. The Hiring Assistant keeps one LangGraph thread per job (one per interview in the room);
a review pauses the graph and resumes from the PostgreSQL checkpoint. LangGraphs 2-5 are started by the application, not by the chatbot.
Candidates never see scores, rankings, transcripts or reports.

Figure 2: Detailed Architecture Diagram